Surfaces from Points and from Satellites

ECON 895 Spatial Techniques in Empirical Economics

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Plan for tonight

Measuring something everywhere

Interpolation

Interpolating with Claude

Night lights

Lights with Claude

Your paper

We break around 8:45. Claude Code does the computing tonight. Have it installed, pull the course project, and install the `gstat` package.

Exercise 2

- Exercise 2 is on Canvas and in the course repository tonight. It is due by email at 7:20 PM on Monday 2 November.
- It has the same two halves as Exercise 1. The build half interpolates a surface and sums lights around cities. The audit half is a script with five planted errors.
- Everything it needs is in `data/` after tonight's pull.

Measuring something everywhere

Two ways to get a value for every place

- Week 4's potato raster had a value in every cell because a crop model computed every cell. Tonight's second half uses a raster a satellite measured, nighttime light.

- Most historical variables were observed at a few places. Temperature came from tree rings and diaries at scattered sites. Plague mortality is recorded for some cities. Getting a value at every city means filling the gaps from the points, which is interpolation.

- Either way the variable in the regression is a model's output. Its errors are correlated across nearby places and they affect both the standard errors and the coefficients.

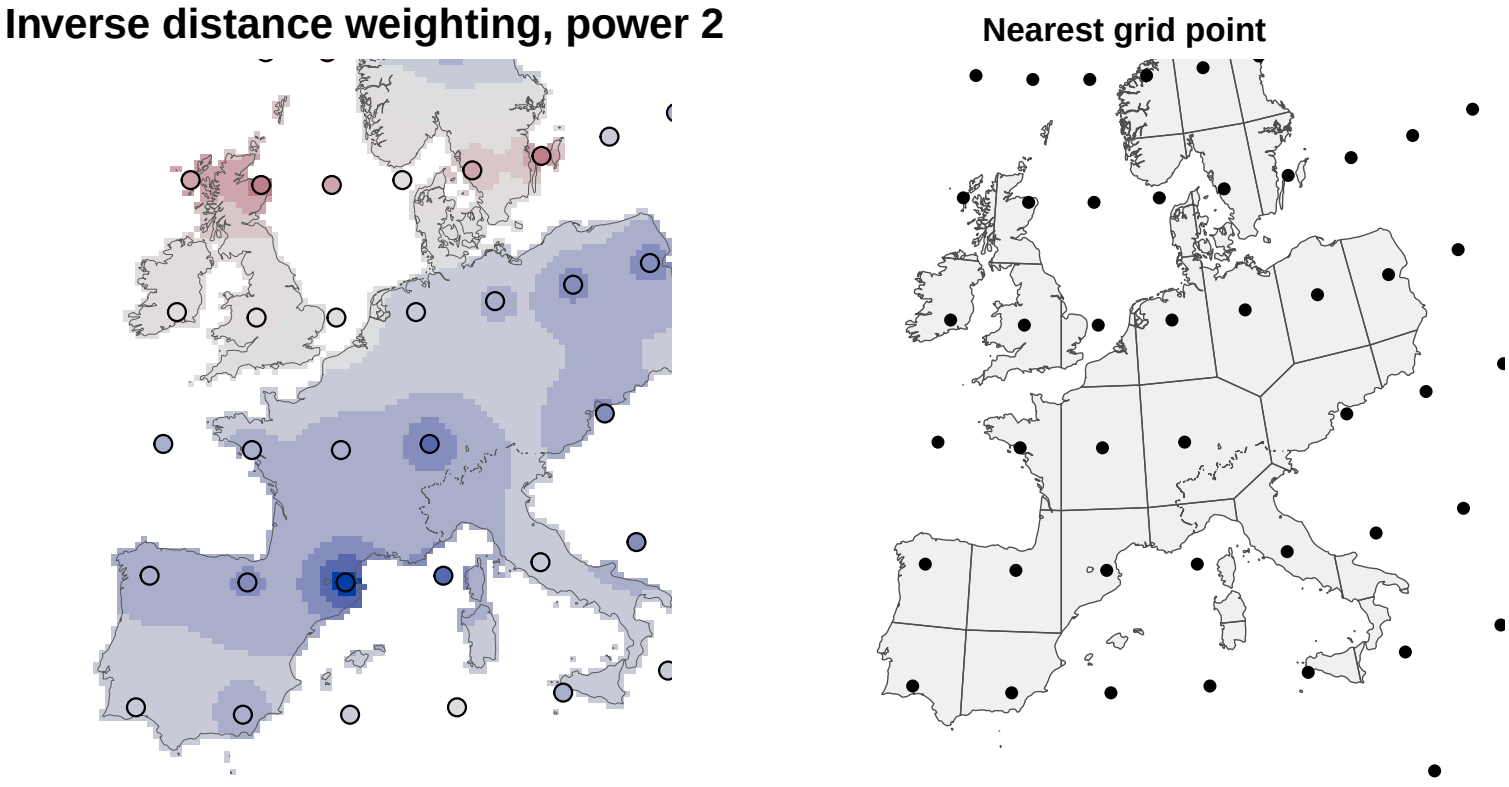
Colder growing seasons raised the probability of a persecution

Anderson et al. (2017) use 1,366 persecutions of Jewish communities in 936 European cities between 1100 and 1800.

- The regressor is the average growing-season temperature in the five years before each period, as a deviation from the 1961–1990 mean, at each city.
- A window one standard deviation colder, about a third of a degree, raised the probability of a persecution in the next five years by 1–1.5 percentage points. The baseline probability was 2 percent.
- The effect was largest where states were weak and soils were poor. The paper attributes part of the long decline in persecutions to market integration and state capacity.

The temperature at each city is interpolated from reconstructed points.

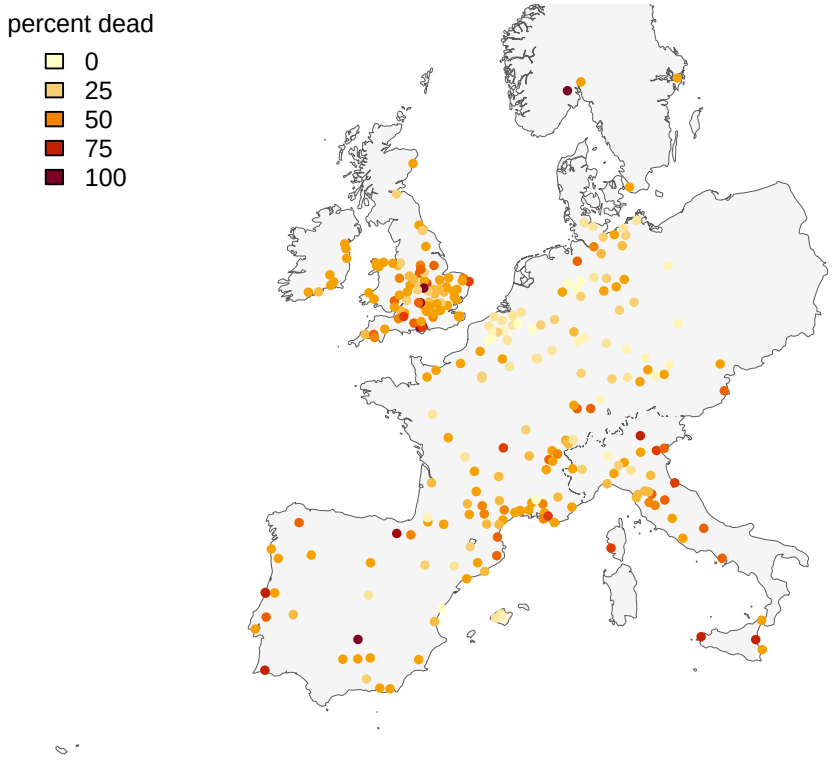
The temperature comes from a grid of reconstructed points



Guiot and Corona (2010) reconstruct April–September temperature from tree rings, documents, ice cores, and pollen on a 5-degree grid of 125 points in the NOAA archive. Left, the average anomaly for 1610–1614 at each point, in degrees Celsius against the 1961–1990 mean, with a surface interpolated between them. Right, the regions closest to each point. Anderson et al. (2017) interpolate to their 936 cities from the 32-point grid they describe and clusters standard errors on the grid point.

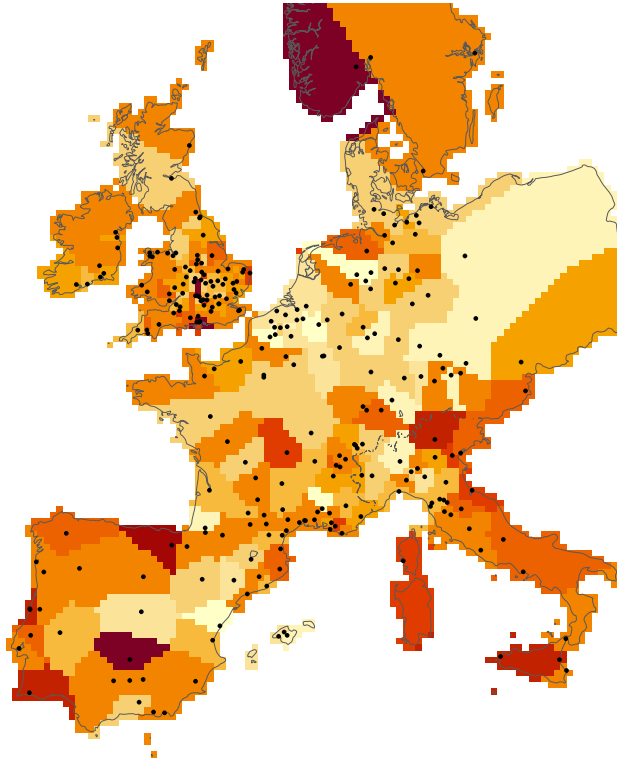
Interpolation

Black Death mortality is recorded for 274 cities



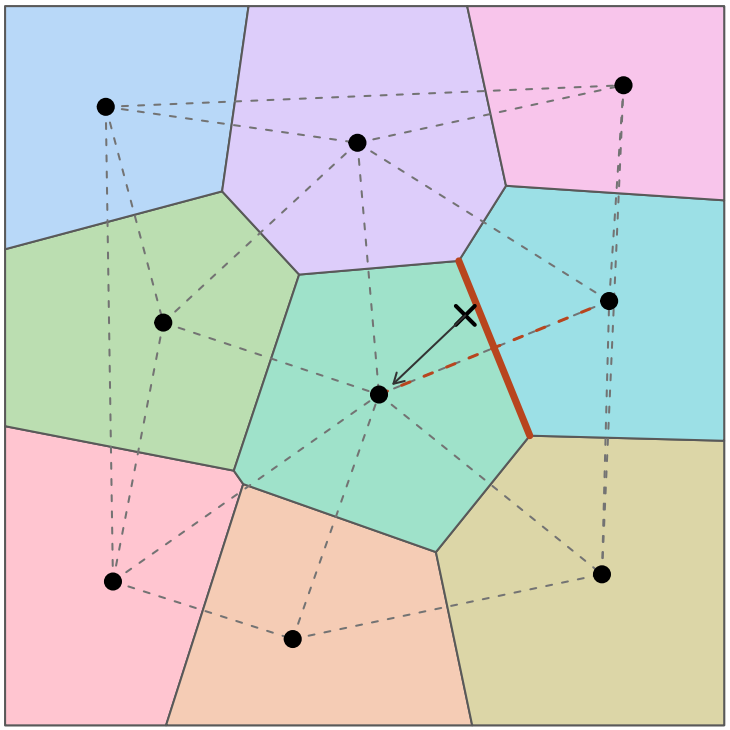
Jedwab et al. (2019) collect the share of the population that died in 1347–1352 for 274 cities. The median is 45 percent and the standard deviation 18.3 points. The full city list has 1,797 cities. The median city on it is 45 km from the nearest city with a recorded rate.

Nearest point gives every city the value of its closest neighbor



Each cell takes the mortality of the nearest recorded city. The surface is flat inside each region and jumps at the borders. The regions are Voronoi polygons, the same construction as the right panel of the temperature map. The method uses one observation per prediction and ignores the rest.

A Voronoi cell is the area closer to one point than to any other



- For points $p_1, \ldots, p_n$ the cell of p_i is $V_i = \{x : d(x, p_i) \leq d(x, p_j)$ for every $j\}$.
- Each edge lies on the perpendicular bisector of two neighboring points. The red edge bisects the red dashed line. A vertex is equidistant from three points.
- The cells are convex and cover the plane without overlap. Outer cells extend without limit and are clipped to the study area.
- Joining points whose cells share an edge gives the Delaunay triangulation, dashed in grey.
- Distance is straight-line in the plane, so project the points first. The cross takes the value of the point its arrow reaches.

When is a Voronoi polygon the right tool?

- **Nearest-point interpolation.** Giving each location the value of its closest observation is the same as filling each Voronoi cell with its point's value. Meteorologists call the cells Thiessen polygons and used them to average rainfall over gauges spread unevenly across a basin.

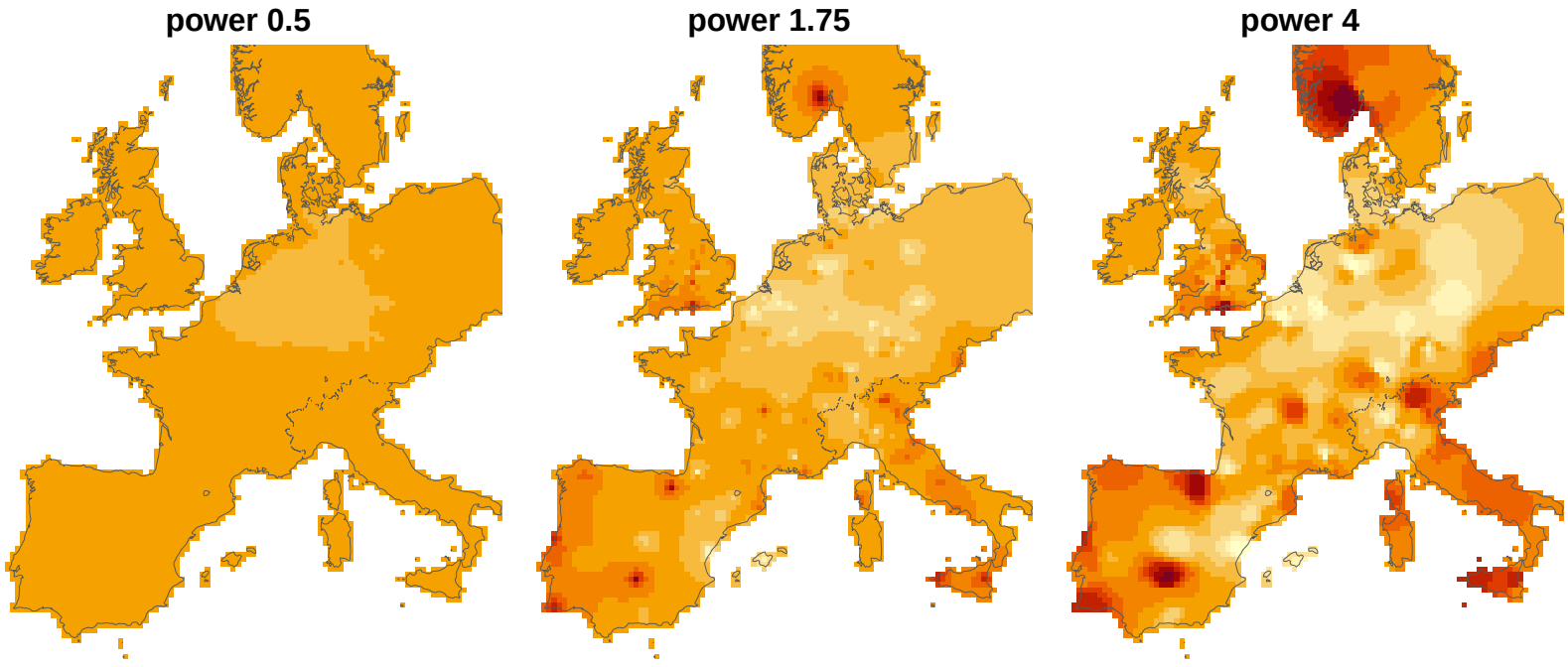
- **Area weights.** A cell's area is the share of the map its point stands for. An average of point data weighted by cell area gives sparse regions their proper weight.

- **Catchments.** Assigning villages, customers, or grid cells to the nearest market, station, or capital draws Voronoi cells around the centers. Each cell is a catchment if people go to the closest center.

- **Grouping.** Clustering cities by their nearest grid point, as Anderson et al. (2017) do, groups them by Voronoi cell.

- **Limits.** The cells use straight-line distance, and Week 7's least-cost distances would draw different ones. Every point counts equally, so a capital and a village get cells by geometry alone. Cells on the edge of the map take their size from where the map was cut.

A higher power gives nearer neighbors more of the weight



- At power 0.5 distant cities still count, so each city's value is close to a continental average and the map is flat.
- At power 4 the nearest recorded city decides almost everything, and the map breaks into the patches of the nearest-point map.
- Power 1.75 sits between them. Leave-one-out cross-validation chooses it.

Weight of a city 50 km away relative to one 200 km away	
power 0.5	2
power 1.75	11
power 4	256

The inverse distance weighting estimator

$$\hat{m}(s) = \frac{\sum_{i=1}^{n} w_i(s)\, m_i}{\sum_{i=1}^{n} w_i(s)}, \qquad w_i(s) = d(s, s_i)^{-p}$$

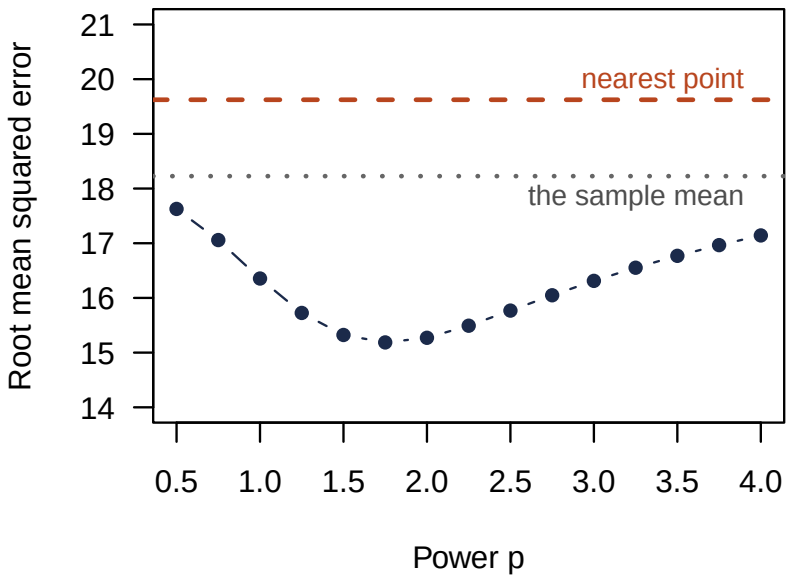
- s is the city to predict, s_i is recorded city i, m_i is its mortality, and d is the distance between them in kilometers. Here $n = 274$.
- Each weight is the distance raised to the power $-p$, so it shrinks as the city gets farther away. Dividing by the sum of the weights makes them add to one.
- The prediction is therefore a weighted average of recorded values. It always lies between the lowest and highest recorded rate.
- At a recorded city the distance is 0 and its weight is infinite, so the method returns the recorded value exactly.
- As p goes to 0 every weight equals 1 and the prediction is the sample mean. As p grows the nearest city takes all the weight and the prediction is the nearest-point value.

A city with three recorded neighbors.

		weight	
km	rate	p=0.5	p=2
50	60	2.0	16
100	40	1.4	4
200	20	1.0	1
prediction		44.5	54.3

At $p = 2$ the nearest city carries 16 of 21 parts of the weight.

Leave one out to choose the power



Drop one recorded city, predict it from the other 273, and record the miss. Repeat for every city and every power. Then

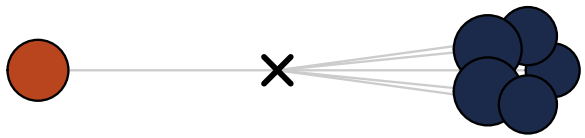
$$\text{RMSE}(p) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(m_i - \hat{m}_{-i}(p)\right)^2}$$

where $\hat{m}_{-i}(p)$ is the prediction for city i made without it. Root mean squared error squares each miss, averages the squares over the $n = 274$ cities, and takes the square root, which returns it to percentage points of mortality. Squaring makes one miss of 40 count as much as four misses of 20.

The error is smallest at $p = 1.75$, where a typical miss is 15.2 points. Nearest point scores 19.6, worse than giving every city the sample mean at 18.2. Recorded cities sit closer to one another than the cities being filled in, with median nearest-neighbor distances of 38 and 53 km, so the error elsewhere on the map is larger.

Why consider kriging instead of inverse distance weighting?

Inverse distance weights, power 1.75



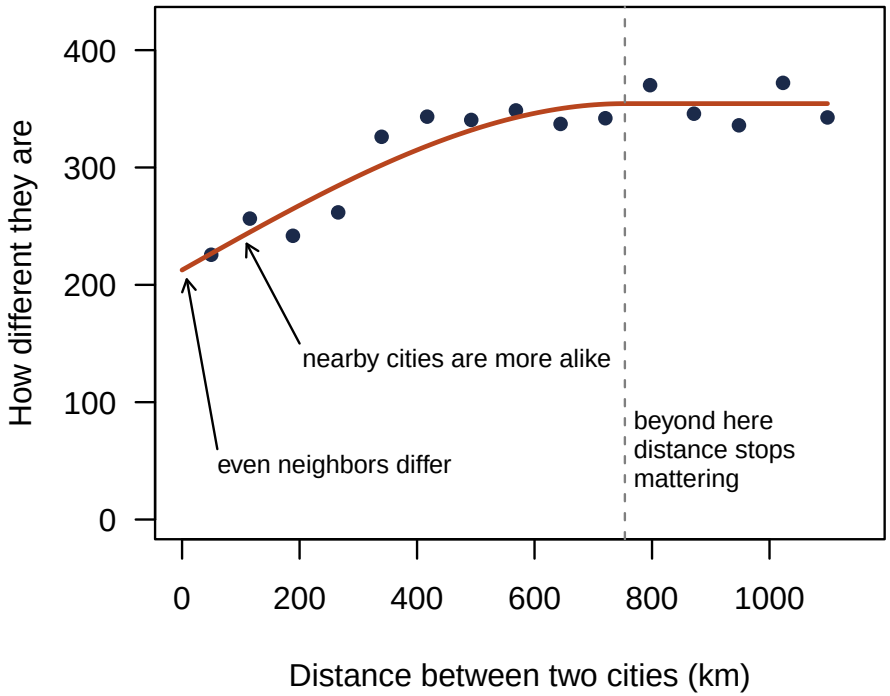
one alone: 16%

five bunched: 84%

Predicting the cross from six recorded cities, all about 100 km away. Each circle's area is its weight.

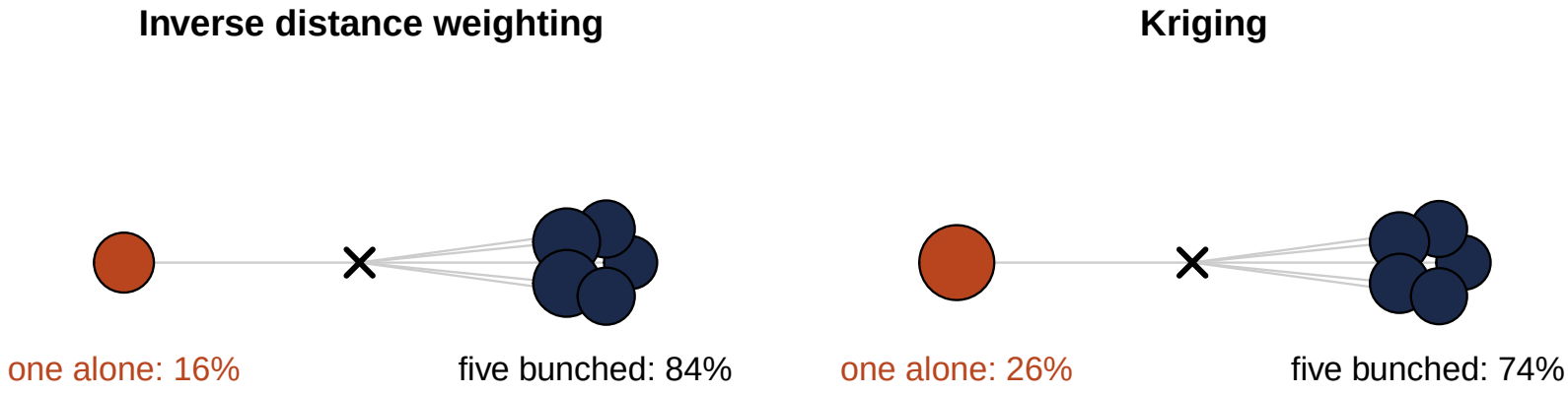
- Inverse distance weighting looks only at how far each recorded city is from the place being predicted. Five cities bunched together get 84 percent of the weight, though they report much the same thing.
- Its power is one number chosen by trial and applied everywhere.
- It gives a prediction with no statement of how reliable that prediction is.
- Kriging learns how quickly similarity fades from the data, discounts bunched cities, and reports a standard error for every prediction.

Kriging first measures how similarity fades with distance



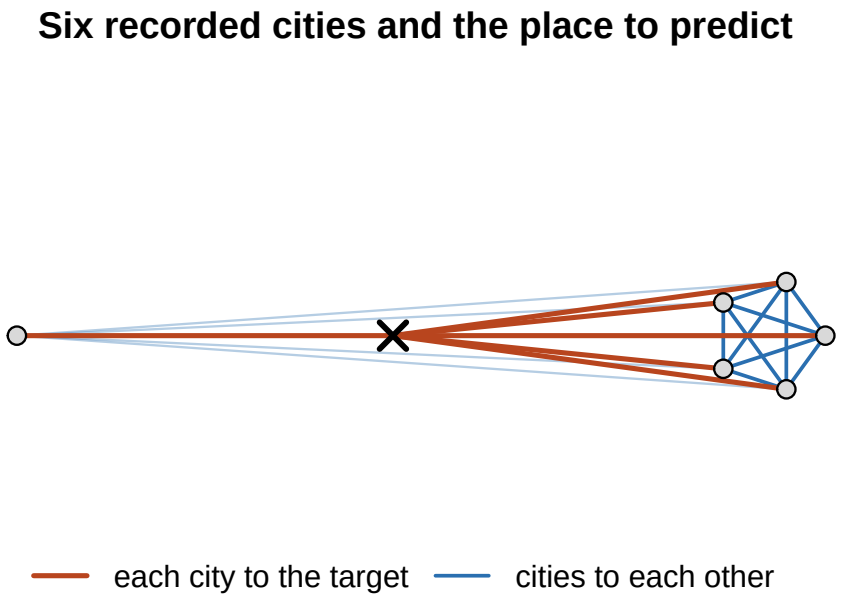
- Take every pair of recorded cities. Plot how different their mortality is against how far apart they are. Each dot averages the pairs at about that distance.
- The dots rise from left to right. Cities close together had more similar mortality.
- The red curve is a smooth shape fitted through the dots. It flattens at about 754 km.
- This plot is called a semivariogram.

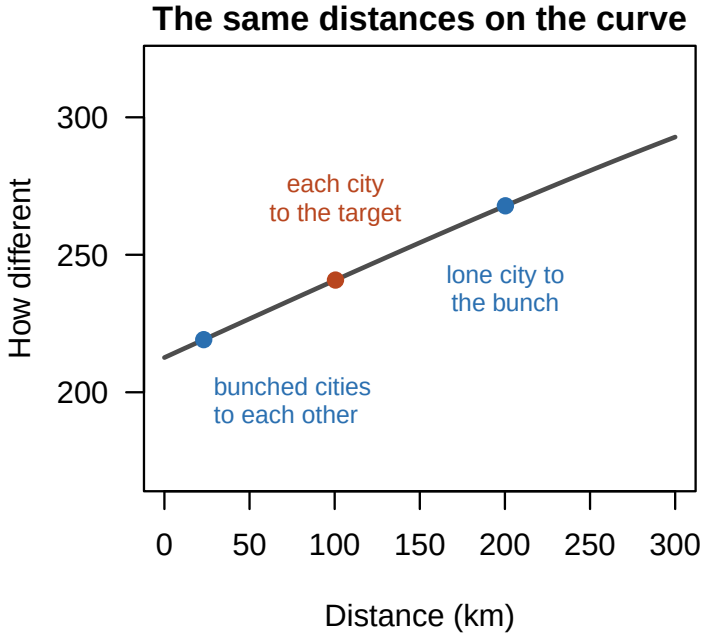
Kriging then sets the weights from that curve



- Kriging reads from the curve how alike two cities at a given distance should be. It applies this both to each recorded city and the place being predicted, and to the recorded cities among themselves.
- Cities bunched together are close to one another, so the curve says they carry overlapping information. They share their weight, and the lone city gains.
- The prediction is still a weighted average of recorded values. Only the weights change.
- The size of the shift depends on the curve. Ours starts high, since even neighboring cities differ, so the cluster keeps most of its weight.

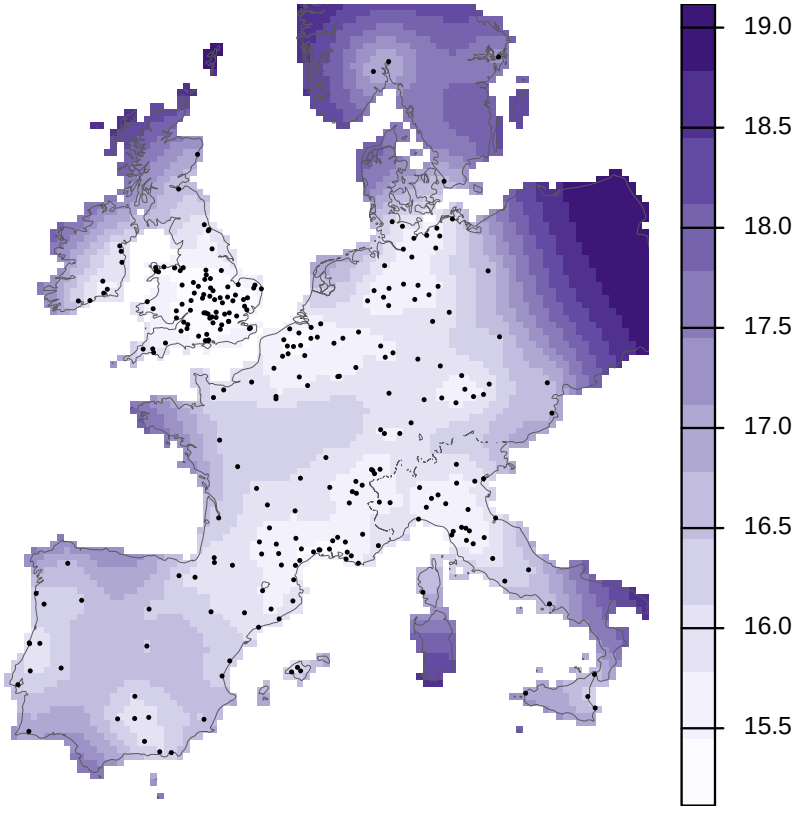
The weights come from two kinds of distance read off the curve





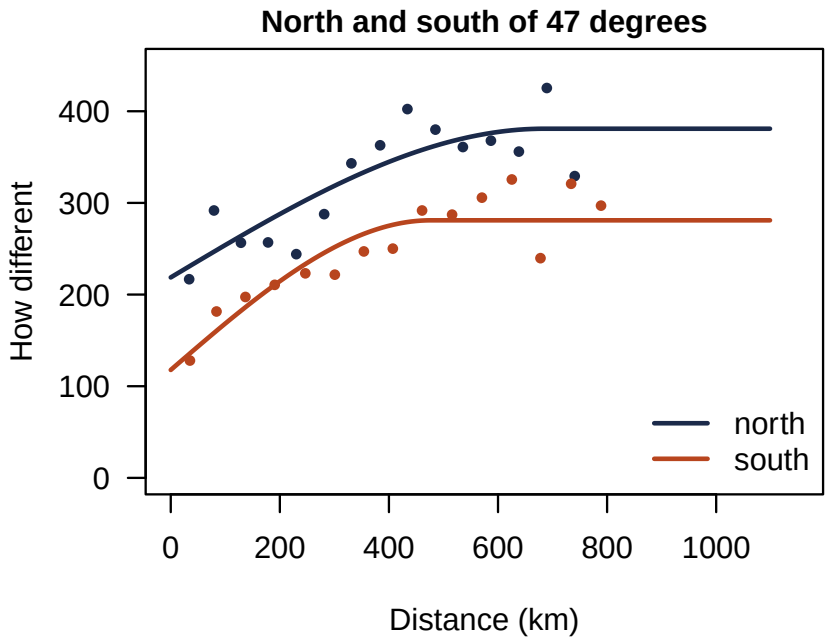
- **Red, each city to the target.** All six sit about 100 km away, so the curve gives each the same expected difference from the place being predicted. On their own, the red lines would split the weight evenly.
- **Blue, cities to each other.** The bunched five are 18–29 km apart, low on the curve, so they are expected to be alike and their errors move together. The lone city is about 200 km from them, higher on the curve, so its error is more independent.
- Kriging chooses the weights that make the expected squared error of the average smallest. Errors that move together do not cancel, so the five bunched cities share their weight and the lone city's weight rises from 16 to 26 percent.

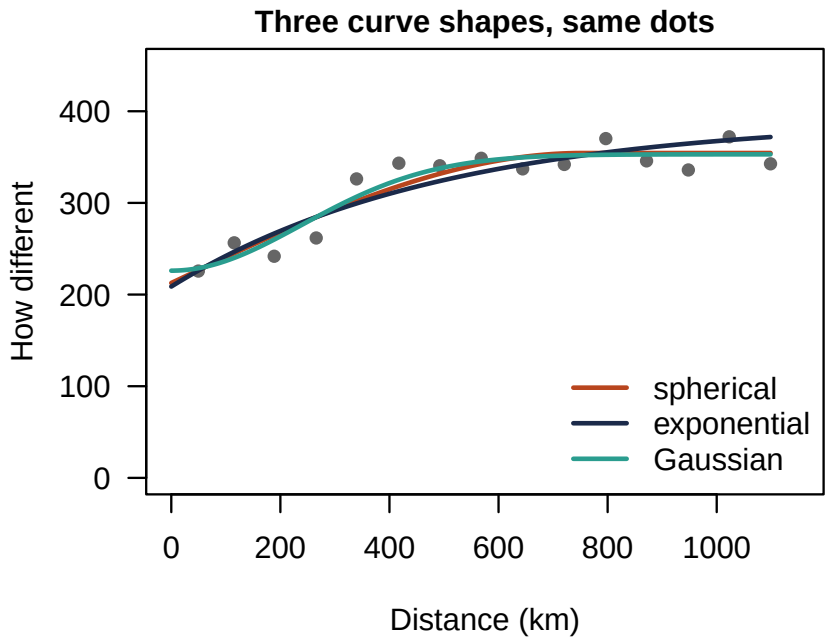
Kriging also reports how uncertain each prediction is



- The same curve tells kriging how far off each prediction is likely to be. The map shows that standard error in points of mortality.
- It is smallest near recorded cities, 15.1, and largest where records are thin, 19.1 in the northeast.
- Even next to a recorded city it stays near 15, because neighboring cities differ by that much.
- Leave-one-out gives kriging an error of 15.5, against 15.2 for inverse distance.

One curve has to describe all of Europe



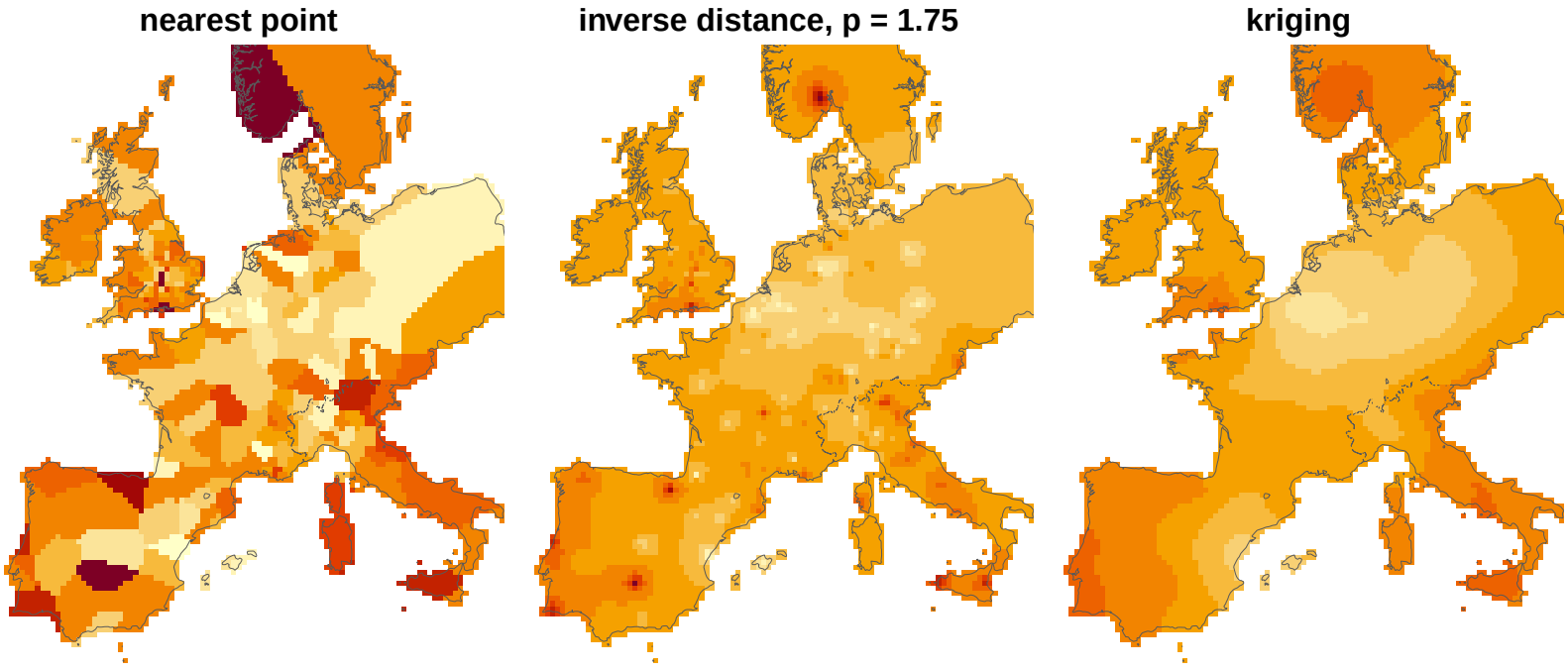


- Kriging assumes similarity fades the same way everywhere. Fitted separately, the north and the south give different curves, and one European curve is a compromise between them.
- The curve's shape is the researcher's choice. Here three standard shapes fit the dots almost identically. With fewer or noisier points they can disagree, and the choice should be reported.

What kriging cannot fix

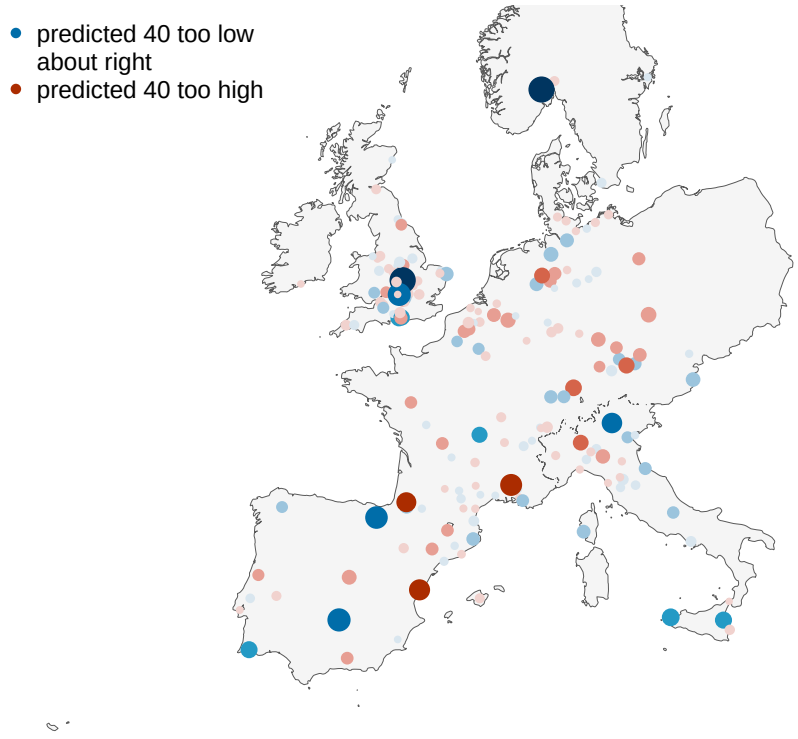
- **Its standard errors are too small.** They treat the fitted curve as known, when the curve was itself estimated from the same 274 points.
- **It smooths.** Like every weighted average, it pulls extreme cities toward their neighbors, so the map varies less than mortality did. As a regressor that smoothed map carries less of each city's own shock.
- **It inherits the points.** If records survive more often from some kinds of places, the surface and its standard errors carry that selection without showing it.
- **Distance is straight-line.** Mountains, seas, and roads do not enter unless you build them in.
- **The gain can be small.** Here kriging and inverse distance weighting predict about equally well. The standard error map is the main thing kriging adds.

The three surfaces



The same 274 points under three rules. Inverse distance and kriging agree on the broad pattern. Inverse distance and nearest point differ by more than 20 points of mortality at 203 of the 1,797 cities.

Where the interpolation misses



Each recorded city, colored by the leave-one-out miss at $p = 1.75$ and sized by its absolute value. The big misses sit where neighboring cities disagree so no method that averages neighbors can recover them. For the regression, that is measurement error in the regressor and it is largest in those regions.

Interpolating with Claude

Starting the project in Claude Code

Copy `examples/surfaces_project/` to `~/wk08-demo`, open a terminal there, and start `claude`. The folder holds the data and a `CLAUDE.md` that specifies the two-part construction in words. It has no scripts.

The request

```
Read CLAUDE.md. Write scripts/01_interpolate_mortality.R to do part 1 of the
construction and run it. Report the leave-one-out root mean squared error at each
power, the chosen power, the error of the nearest-point method and of predicting
the mean, the number of cities where inverse distance and nearest point differ by
more than 20 points, and the interpolated mortality for LYON, STRASBOURG, LEIPZIG,
SEVILLA, and EDINBURGH.
```

Read the script it writes before reading any number it reports. Find the line that projects the points and the line that leaves one out.

Expected output for part 1

These numbers come from the same construction, run when the slides were built. Claude's report should match them.

chosen power	1.75
error at that power	15.2
error at power 0.5	17.6
error at power 4	17.1
nearest point	19.6
sample mean	18.2
cities differing by more than 20	203

	inverse	nearest
Lyon	35	35
Strasbourg	37	33
Leipzig	35	33
Sevilla	50	50
Edinburgh	33	33

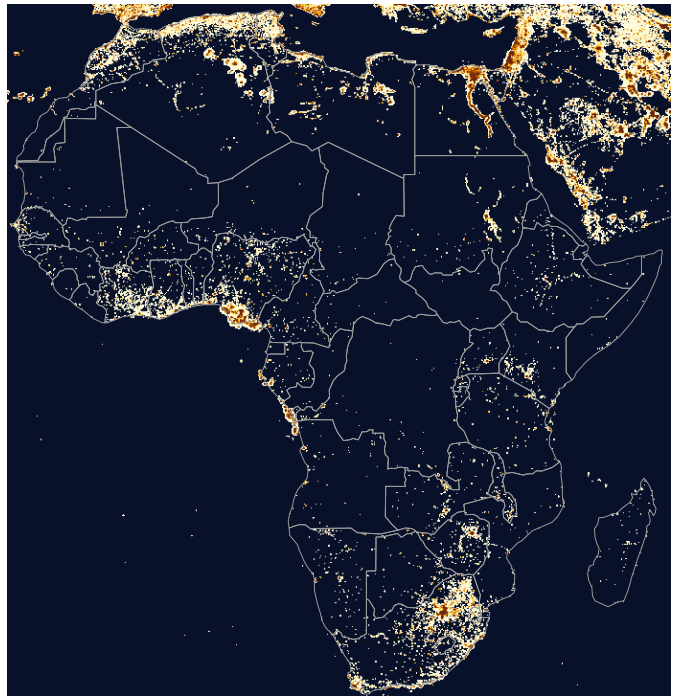
If a number differs find the line in the script that produces it. The usual departure is leaving the predicted city in its own training set, which drives every error to zero and picks the smallest power on the grid.

Break

Ten minutes.

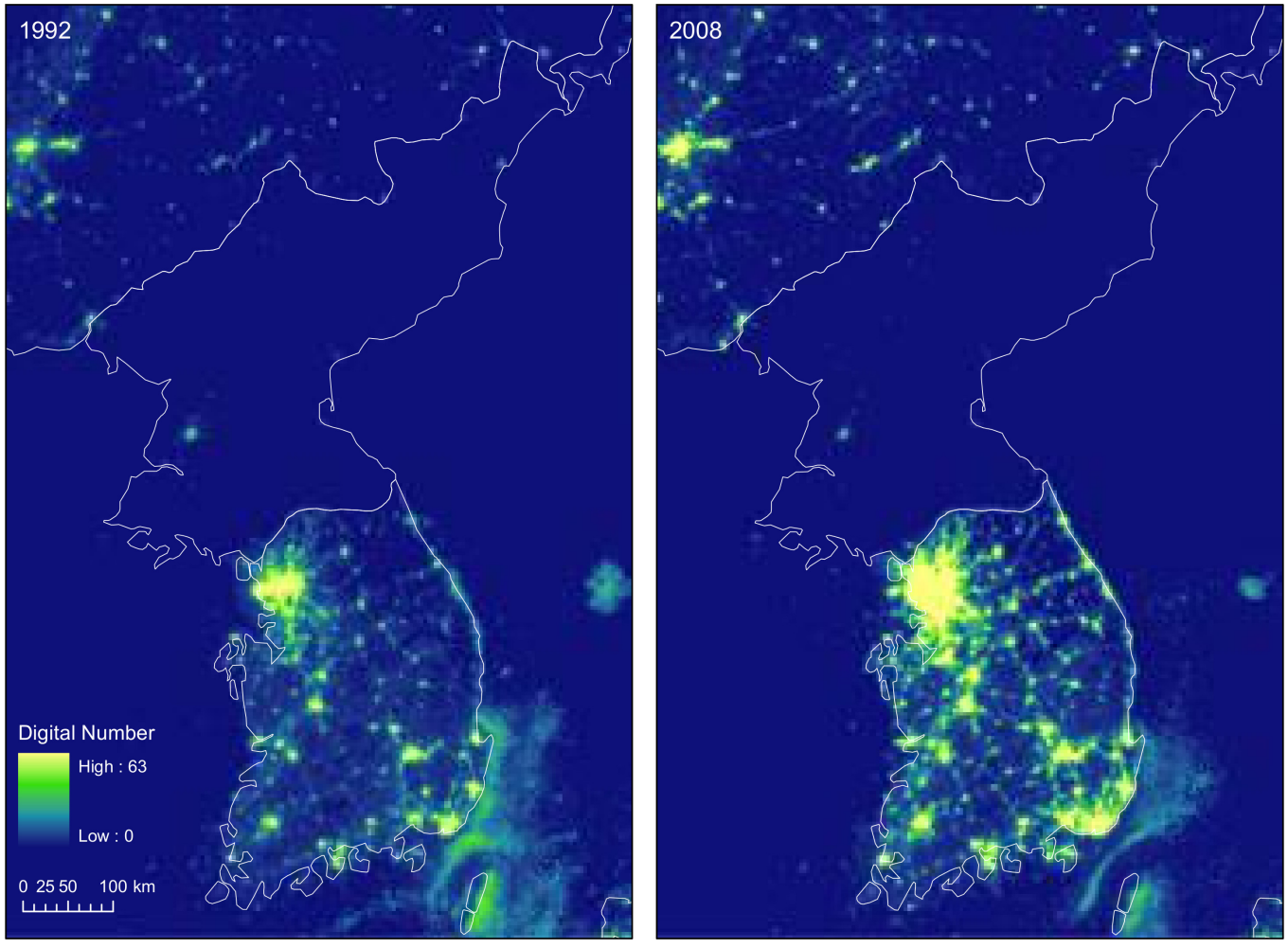
Night lights

The DMSP stable lights composite



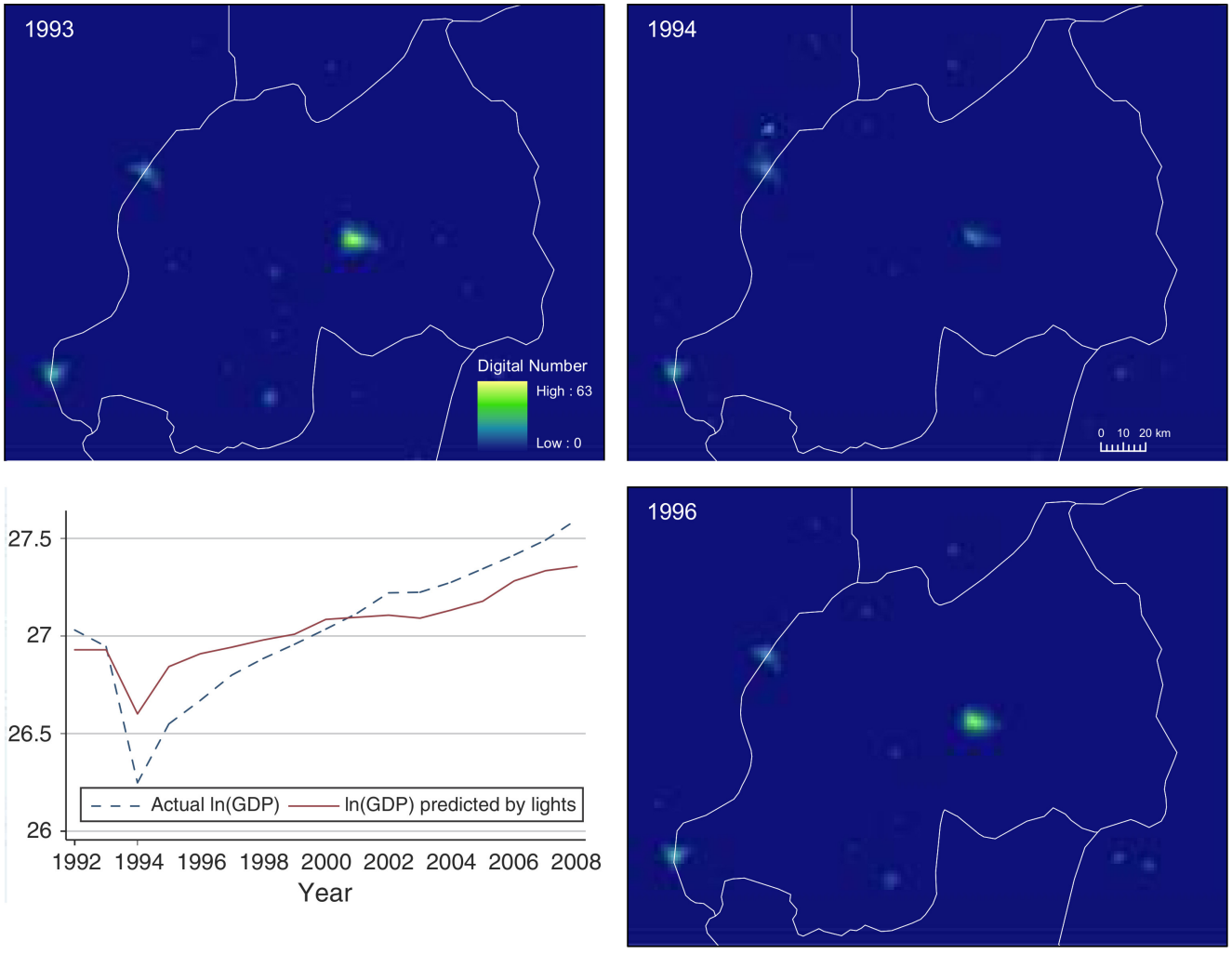
The Defense Meteorological Satellite Program's Operational Linescan System recorded the Earth at night, and NOAA's digital composites cover 1992–2013. NOAA removed clouds, fires, and moonlight and averaged a year's cloud-free nights into one composite of stable lights. Each cell is 30 arc-seconds, about a kilometer, and holds a digital number from 0 to 63. This is Africa in 2000, aggregated to 5 arc-minutes by taking each cell's brightest kilometer. Of the land cells, 2.9 percent have any light. The sensor has no onboard calibration, so values differ across satellites and years. VIIRS replaced it from 2012.

The two Koreas, 1992 and 2008



Stable lights, digital number 0–63. From the 2022 version of this lecture.

Rwanda, 1993–1996



Lights around Kigali before, during, and after the 1994 genocide, with the national accounts beside them. From the 2022 version of this lecture.

Lights track income where income is badly measured

- Henderson et al. (2012) show that within a country, growth in measured light moves with growth in GDP. They use the relation to produce growth estimates for countries whose national accounts are poor and to measure growth for regions that have no accounts at all.

- The unit of observation can be anything with a boundary, such as a country, a province, an ethnic homeland, a 10 km circle around a city, or a cell. Sum the digital numbers inside it, as Week 4 summed potato suitability.

- Divide by population from a gridded source such as the Gridded Population of the World, which reports people per square kilometer in each cell.

Turning population density into a count of people

Light per person needs a count of people. The population raster stores density.

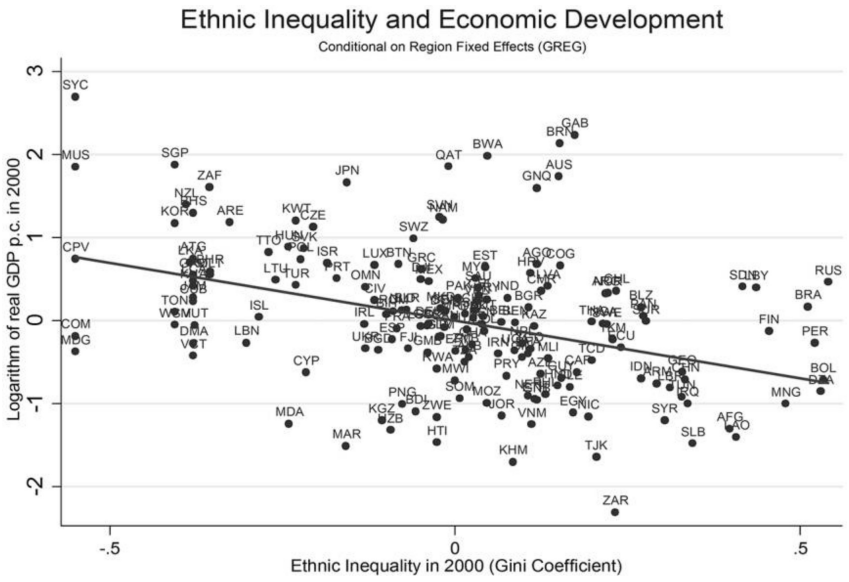


One cell, outlined in blue. 10,203 people per km^2 $\times$ 21.4 km^2 $\approx$ 218,000 people.

Each of the 16 cells times its area, summed	1.36 million people
The 16 densities summed without the area	63,530, not a count

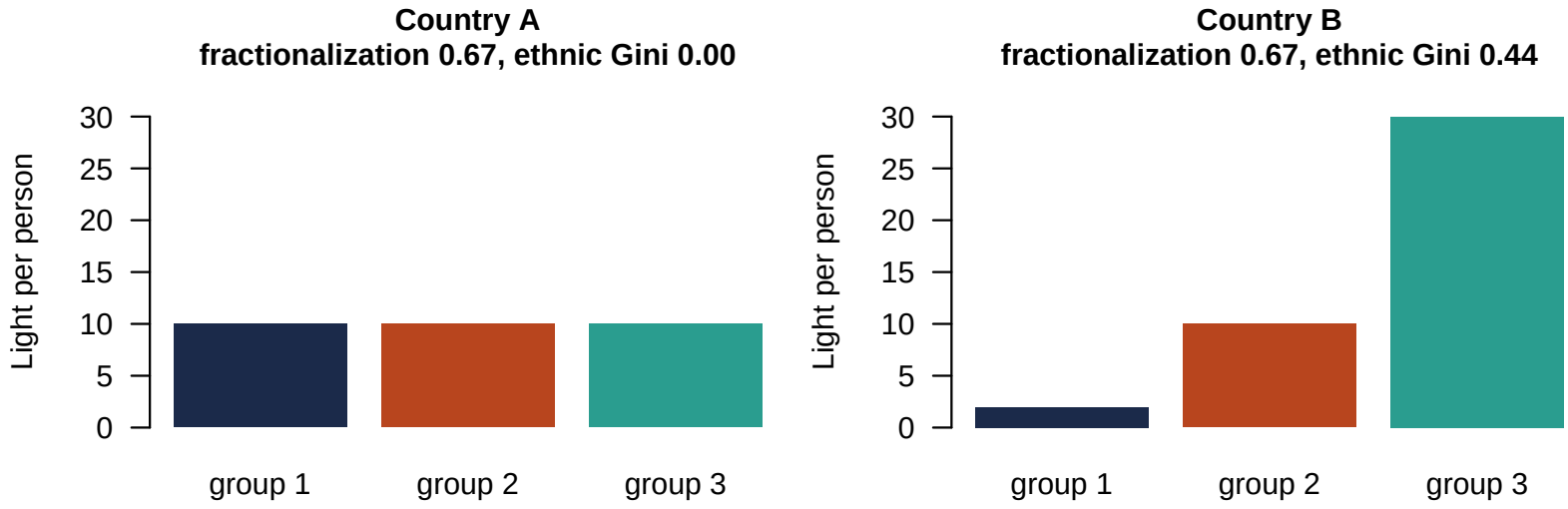
Cells shrink away from the equator, so each needs its own area. Over Cairo the same grid's cells cover 18.6 km^2.

Ethnic inequality from lights



- Alesina et al. (2016) define ethnic inequality as the gap in living standards between a country's ethnic groups.
- They compute light per person in each group's homeland from an ethnolinguistic atlas and take a Gini coefficient across the groups, each counted once whatever its size. A Gini of 0 means every homeland is equally lit per person.
- Countries with higher ethnic inequality are poorer, controlling for spatial inequality measured without reference to ethnicity.
- Differences in the geography of the homelands explain a large share of the gap between groups.

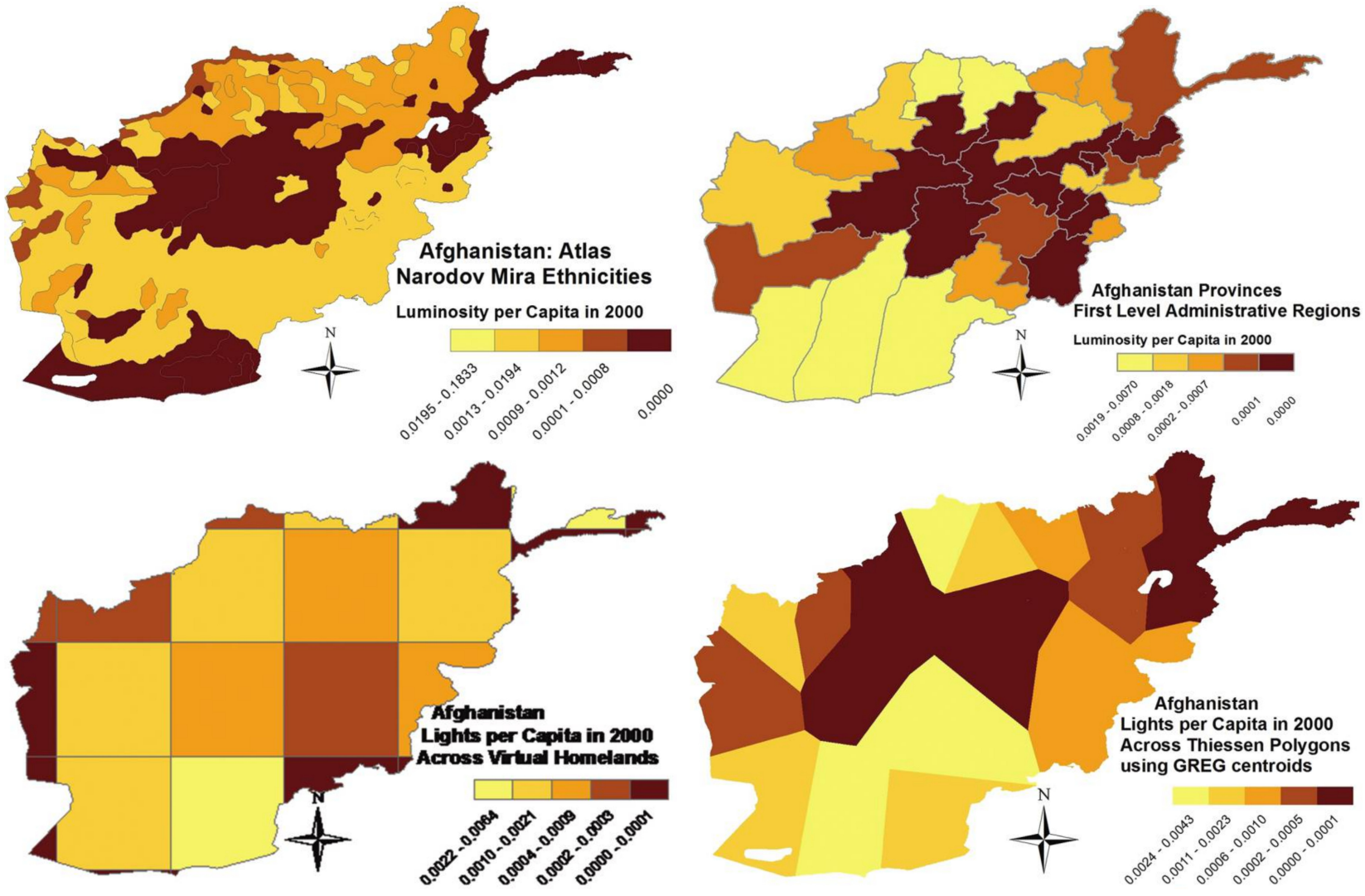
Ethnic inequality and ethnic fractionalization measure different things



Both countries have three groups of equal size.

- **Fractionalization** is $1 - \sum_g s_g^2$, where s_g is group g's population share. It is the chance that two people drawn at random belong to different groups. It uses only the group sizes, so A and B score the same.
- **Ethnic inequality** uses only how well off each group is. A scores 0 and B scores 0.44.
- The two are modestly correlated across countries. Fractionalization is associated with lower GDP per capita, and in Alesina et al. (2016) that association becomes statistically indistinguishable from zero once ethnic inequality is controlled for.

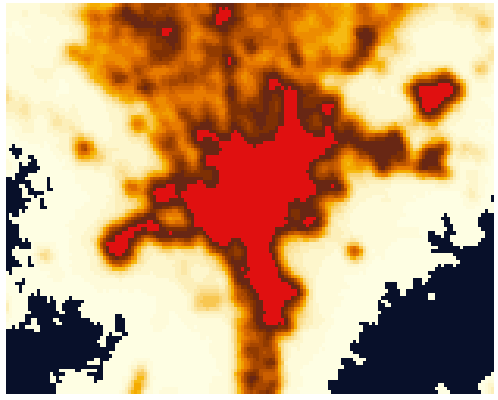
The same lights under four partitions of Afghanistan



Light per person in 2000 by ethnic homeland, by province, by 2.5-degree cell, and by Voronoi polygon around homeland centroids. A Gini computed across units depends on the units. From the 2022 replication of Alesina et al. (2016).

Four limits of the sensor

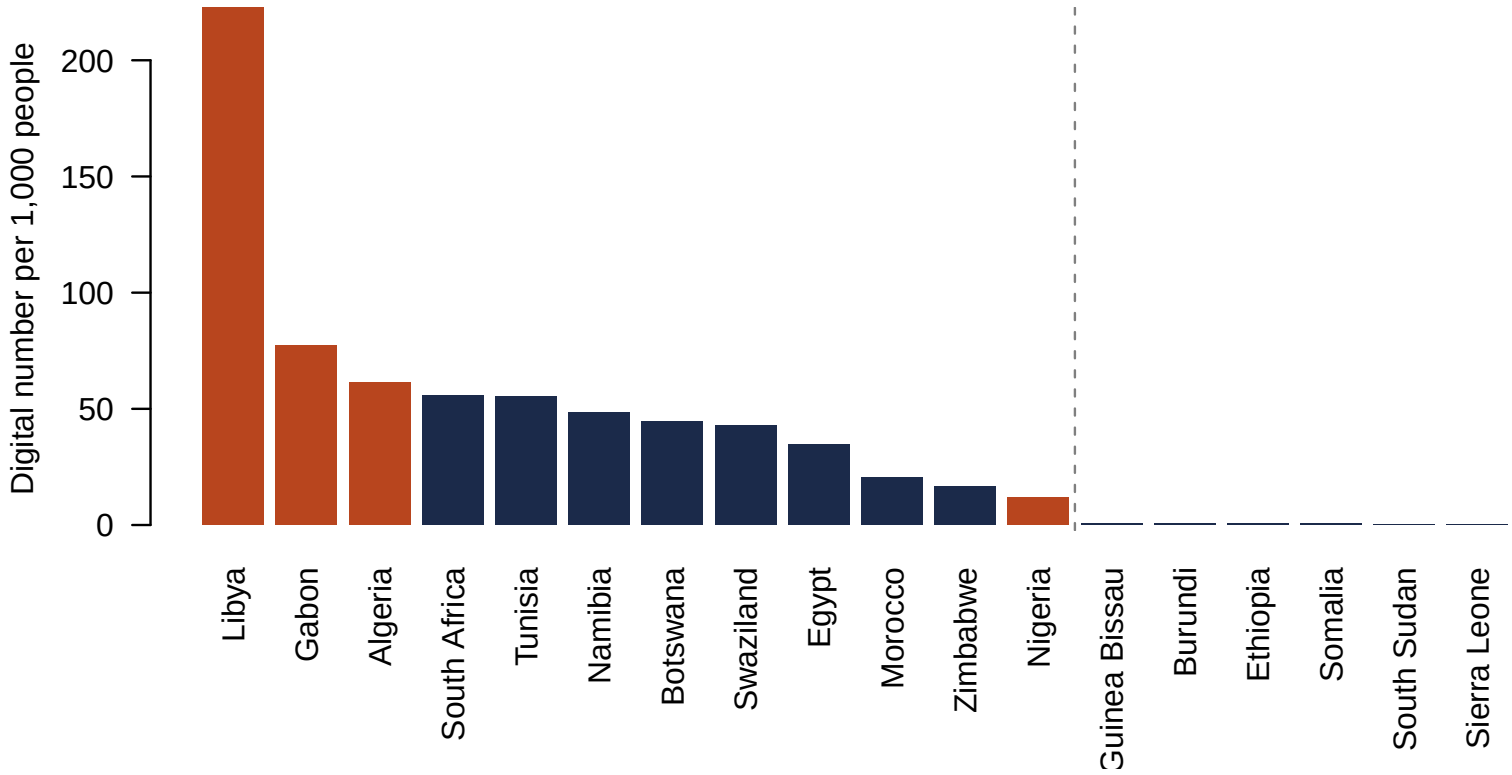
Cairo and the Nile Delta



Red cells are at 63, the ceiling.

- **Top-coding.** The sensor saturates at 63, where 8,731 African cells sit and hold 4.5 percent of the continent's summed light. Central Cairo and central Lagos both read 63.
- **Zeros.** Of the land cells, 97.1 percent read 0. Every cell below the detection floor reads 0, whatever its population or income.
- **Blooming.** Scattered light makes a lit area look wider than it is and spreads coastal light onto the sea.
- **Flares.** Gas flares at oil fields are bright and have no people under them.

Light per person by country



The 12 highest and six lowest of 51 countries and territories on present-day borders, with lights for 2000. Libya, Gabon, Algeria, and Nigeria, in rust, are oil producers whose gas flares add light where few people live. Light per person ranks the first three above South Africa, and the flares account for part of that ranking.

Lights with Claude

The request for part 2

The request

```
Write scripts/02_lights_by_country.R to do part 2 of the construction and run it.
Report the five countries with the most light per 1,000 people and the five with
the least, the share of all of Africa's summed digital number that sits in cells
at 63, and the share of land cells with any light.
```

Before it runs say what the population step has to do. It multiplies density by cell area and sums the result. A script that sums density on the native grid gets a population about 20 times too small because each 2.5-arc-minute cell covers about 20 square kilometers. A script that first resamples density to the 1 km lights grid and then sums it is off by that grid's cell area instead and comes out about 20 percent too large, because each 30-arc-second cell covers less than one square kilometer. Either way every per-person figure is wrong by that factor.

Expected output for part 2

Most light per 1,000 people	
Libya	223
Gabon	77
Algeria	61
South Africa	56
Tunisia	55
Least	
Sierra Leone	0.2
South Sudan	0.4
Somalia	0.6
Ethiopia	0.8
Burundi	0.8

light in cells at 63	4.5%
land cells lit	2.9%
summed digital number, millions	12.3

The per-person figures depend on the population step. If Claude's values are off from these by a roughly constant factor check whether it multiplied density by cell area. That error scales every country alike and leaves the ranking almost unchanged.

Your paper

Which surface does your question need?

- A variable you observed at points and need at your units requires interpolation. Choose the method, pick its parameter by leaving points out, report the error, and cluster on the source points.

- A variable a satellite measured is a raster. Sum or average it inside your units with correct area weights and know the sensor's floor and ceiling before you compare two places.

- The interpolated value is a prediction from other places and the raster value is a proxy for the variable you want.

Before next week

Exercise 2 is due by email at 7:20 PM on Monday 2 November as three files named with your last name.

Week 9

- Week 9 is cities and persistence, with Davis and Weinstein (2002) on Japanese cities after the bombing and Bleakley and Lin (2012) on portage sites.

- Week 9 also covers spatial standard errors. Conley and Kelly (2025) reanalyze the portage result and ask how much of persistence is spatial noise. Clustering on the grid point, as Anderson et al. (2017) do, addresses the same problem of errors correlated across nearby places.

References

Alesina, A., Michalopoulos, S., and Papaioannou, E. (2016). Ethnic inequality. *Journal of Political Economy*, 124(2):428–488.

Anderson, R. W., Johnson, N. D., and Koyama, M. (2017). Jewish persecutions and weather shocks: 1100–1800. *The Economic Journal*, 127(602):924–958.

Bleakley, H. and Lin, J. (2012). Portage and path dependence. *The Quarterly Journal of Economics*, 127(2):587–644.

Conley, T. G. and Kelly, M. (2025). The standard errors of persistence. *Journal of International Economics*, 153:104027.

Davis, D. R. and Weinstein, D. E. (2002). Bones, Bombs, and Break Points: The Geography of Economic Activity. *American Economic Review*, 92(5):1269–1289.

Guiot, J. and Corona, C. (2010). Growing season temperatures in Europe and climate forcings over the past 1400 years. *PLoS ONE*, 5(4):e9972.

Henderson, J. V., Storeygard, A., and Weil, D. N. (2012). Measuring economic growth from outer space. *American Economic Review*, 102(2):994–1028.

Jedwab, R., Johnson, N. D., and Koyama, M. (2019). Negative shocks and mass persecutions: Evidence from the Black Death. *Journal of Economic Growth*, 24(4):345–395.